

Stock returns and output growth in emerging and advanced economies

Paolo Mauro

Research Department, International Monetary Fund, 700 19th Street NW, Washington, DC 20431, USA

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Abstract

This paper studies the correlation between output growth and lagged stock returns in a panel of emerging market economies and advanced economies. It finds that the proportion of countries in which this correlation is significant is the same for emerging market economies as it is for advanced economies using yearly data, and somewhat lower using quarterly data. Asset prices therefore seem to contain valuable information to forecast output also in emerging market economies. Moreover, the paper finds that the strength of the correlation between output growth and lagged stock returns is significantly related to a number of stock market characteristics, such as a high market capitalization to GDP ratio and, less robustly, English legal origin and the number of listed domestic companies and initial public offerings.

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There is no sense in building up a new enterprise at a cost greater than that at which a similar existing enterprise can be purchased; whilst there is an inducement to spend on a new project what may seem an extravagant sum, if it can be floated off on the stock exchange at an immediate profit. (J.M. Keynes, *The General Theory of Employment, Interest and Money*, 1936)

Cause and effect run from the economy to the stock market, never the reverse. (John Kenneth Galbraith, 1955)

E-mail address: pmauro@imf.org (P. Mauro).

The stock market has predicted nine out of the last five recessions. (Paul Samuelson, 1960)

1. Introduction

The recent financial crises and associated output declines experienced by a number of emerging market economies have raised anew the issue of the links between financial and real variables, notably between stock market performance and economic activity. The empirical association between real economic activity and lagged real stock returns has been analyzed in depth in the case of the United States and, to a lesser extent, other advanced countries. However, this relationship remains surprisingly unexplored in the case of emerging market economies.¹ Moreover, while a number of theories have been proposed to explain this empirical association, cross-country differences in its strength have not been used to discriminate among existing theories.

To my knowledge, the present paper is the first to analyze systematically the correlation between real stock returns and economic growth in a panel of emerging market and advanced economies. An interesting and extensive literature has already analyzed the relationship between stock market development and long-run economic growth in panels of countries (see, e.g., [Levine, 1997](#); [Levine and Zervos, 1998](#)). The present paper considers a different, if somewhat related, question, and focuses on the short-run relationship between stock returns and output growth.² By doing so, it seeks to fill two gaps in the existing literature. First, it examines the extent of this correlation in emerging countries, and compares it to that observed in advanced countries, to prepare the ground for improved leading indicators of output for emerging countries. Second, it uses this larger group of countries to address the more general question of what type of countries would tend to display a stronger association between output growth and lagged stock returns, and it relates the results to existing theories.

In the United States, the correlation between real economic activity and lagged real stock returns is positive and both statistically and economically significant, as documented by several studies, including [Fischer and Merton \(1984\)](#), [Barro \(1990\)](#), [Fama \(1990\)](#), and [Schwert \(1990\)](#). A similar relationship holds in Canada ([Barro, 1990](#)), Japan, Germany, and the United Kingdom ([Mullins and Wadhwani, 1989](#)), the G-7 ([Choi et al., 1999](#)), and several other European countries ([Aspremi, 1989](#); [Wasserfallen, 1989](#)).

Although the correlation is significant and stock returns provide valuable information about future economic activity, they are far from perfect as a leading indicator, as suggested by Samuelson's oft-quoted adage reported above. [Stock and Watson \(1990\)](#),

¹ More generally, studies on leading indicators in developing countries are relatively scarce. In an exception, [Agenor et al. \(2000\)](#) explore the extent to which a number of macroeconomic variables lead the economic cycle in developing countries.

² The present study was issued as an IMF working paper in May 2000. A related working paper by [Hassapis and Kyprianos \(September 2000\)](#) was brought to my attention by the co-editor and the referee. That paper considers the relationship between stock returns and industrial production growth at the monthly frequency for Chile, Colombia, Mexico, Peru, and the United States. I compare my findings to those of Hassapis and Kyprianos in Section 3.

1998) show that the relationship between stock returns and economic growth has not been stable over time, and that the predictive information of stock returns for future activity is also contained in other financial variables — such as yield spreads between 10-year and 3-month government bonds, or between T-bills and private commercial paper — in the case of the United States.³

Several theories (reviewed in the next section) have been put forward to provide a rationale for this empirically observed correlation. All theories allow for the possibility that news about future output growth is reflected in stock returns. Some authors (including John Kenneth Galbraith, as in the quote reported above) argue that this is the only mechanism underlying the correlation. This mechanism does not involve stock price changes “causing” changes in output in an economic sense, though of course it does involve Granger-causality, with lagged stock returns predicting output growth. Others argue that, in addition, stock price changes that are not justified by fundamentals may cause changes in output, through mechanisms that typically rely on stock markets being sufficiently developed in a number of respects discussed below.

There are three reasons why analyzing the link between stock returns and output growth is especially interesting for emerging markets. First, leading indicators are relatively scarce in emerging markets. In particular, relatively low financial market liquidity and frequent changes in financial structure imply that other financial variables such as yield spreads are unlikely to be successful predictors of output, and in any case, it is usually difficult to identify a relevant yield spread for a sufficiently long sample period. Given the speed with which stock market prices become available, it seems that the stock market could be a helpful leading indicator in forecasting economic growth. Immediately following the onset of the Asian crisis in 1997, there was no consensus on how deep an impact the financial crisis would have on economic growth. At that time, the extent of the decline in stock price indices might have provided useful guidance in answering that question. Second, volatility of returns in emerging markets is higher than in mature markets (Richards, 1996) and the same is true of output. While this might reflect greater noise, it might also yield a wealth of information, even though data on emerging countries are usually available for shorter sample periods than data on advanced countries. Moreover, if one were to find that stock price changes that are not justified by fundamentals really affected output, this would raise a number of policy issues that would be particularly relevant for emerging markets. Third, by expanding the sample of countries to include a number of emerging markets, it is possible systematically to address the question of whether the association between stock returns and output growth is stronger in some types of countries than others, depending not only on their level of economic development, but also on several indicators of the size and liquidity of, and legislation governing, their stock markets.

This paper presents two main findings. First, the empirical association between output growth and lagged stock returns, albeit far from perfect, is significant in several (but by no means all) emerging and advanced countries. The proportion of emerging market countries in which this association is significant is the same as for advanced countries at the yearly frequency, but somewhat smaller at the quarterly frequency. Second, the association is

³ Campbell (1999) analyzes the forecasting power of other financial variables in a number of advanced economies.

significantly stronger in countries that have high market capitalization, a large number of listed domestic companies and initial public offerings, and English (or non-French) origin of the regulations governing the stock market. Although all of these country characteristics are correlated, those with the best (even if far from perfect) predictive power for whether a country has a strong association between output growth and stock returns are market capitalization and — a less robust finding — English (or non-French) legal origin of the regulations governing its stock market.

The remainder of this paper is organized as follows. Section 2 reviews existing theories of the link between stock returns and output growth and relates them to a number of characteristics of countries' stock markets. Section 3 describes the data on stock market returns and economic growth and reports the estimation results. Section 4 concludes.

2. Theories on the link between stock returns and output growth

The observed correlation between economic growth and stock returns has prompted a debate on the causal direction of the underlying relationship. In an excellent survey, [Morck et al. \(1990\)](#) review five existing theories of this link which arguably encompass most of the main theories that have been put forward in this literature. This section draws on [Morck et al. \(1990\)](#) to describe those theories, and briefly summarizes the empirical literature that tests them. It then conjectures which country characteristics may predict the strength of the growth-returns association, according to each theory. The theories may be grouped into those according to which stock price movements not reflecting changes in future “fundamentals” cannot predict changes in output (the “passive informant” hypothesis and the “accurate active informant” hypothesis), and those according to which they can (the “faulty active informant”, the “financing” hypothesis, and the “stock market pressure on managers” hypothesis).

- (1) According to the “passive informant” hypothesis, the only mechanism underlying the correlation between stock returns and output growth is the following. Under the assumptions that stock prices reflect the present discounted value of all future dividends and that dividend growth is related to GDP growth, a correlation between this year's stock returns and next year's economic growth arises naturally: if next year's economic growth is buoyant, news revealed this year will typically be positive, resulting in large stock price increases this year. All theories reviewed below accept that the above mechanism plays a role, but leave room for additional mechanisms.
- (2) Under the “accurate active informant” hypothesis, stock price changes provide managers with information about the market's expectations of future economic developments. Managers base their investment decisions upon that information, thereby justifying the market's expectations. In this case, stock price changes turn out to be perfectly correlated with fundamentals.
- (3) In the “faulty active informant” hypothesis, managers' decisions about investment are influenced by stock price movements, but managers cannot distinguish between movements reflecting fundamentals and those reflecting market “sentiment”. Stock market movements that are not motivated by fundamentals can therefore mislead

Table 1
Implications of various theories for strength of growth-returns links

Theory	Country characteristics					
	Emerging/Advanced or per capita GDP	Market capitalization	Turnover	Initial public offerings	Listed companies	Legal origin
Passive informant	No	Possibly	Possibly	No	Possibly	No
Accurate active informant	No	Possibly	Possibly	No	Possibly	No
Faulty active informant	No	Possibly	Possibly	No	Possibly	No
Financing	No	Yes	Yes	Yes	Yes	No
Stock market pressure	No	Yes	Possibly	No	Yes	Yes

According to each theory, do the above country characteristics — holding other characteristics constant — make it more likely that a country will display a strong association between output growth and lagged stock returns?

managers into overinvesting or underinvesting compared with what later turns out to be warranted by fundamentals.

- (4) The “financing” hypothesis, based upon Tobin’s q theory (a formalization of Keynes’ reasoning in the quote reported above), argues that when stock prices are high compared to the replacement cost of capital, entrepreneurs will be more likely to expand their activities by investing in new physical capital (possibly financed by issuing new shares of their company) rather than by purchasing existing firms on the stock market.
- (5) Finally, the “stock market pressure on managers” hypothesis suggests that stock price changes can affect investment even if they neither convey information nor change financing costs. If investors hold negative views on a firm’s prospects and drive down its stock price, managers may have to cut their investment projects to protect themselves from the possibility of being fired or taken over.

Although lack of data on fundamentals (i.e., dividends) prevents direct testing using the large cross-country data set analyzed in this paper, each of these theories has different implications for the country characteristics that might be related to a strong growth-returns link. In particular, the following conjectures can be made. (These are summarized in Table 1, which reports whether each of the five theories would predict a certain country characteristic to lead to a closer growth-returns association, holding other things equal — including the other characteristics in the table.)

- Under the “passive informant” hypothesis, most country characteristics are unlikely to predict the strength of the association between stock returns and output growth, because good news about output leads to a capital gain on stocks regardless of country characteristics.⁴
- Similarly, under the “active informant” (whether “accurate” or “faulty”) hypothesis, market capitalization might matter, because a larger stock market implies that stock

⁴ One possible exception is the extent to which the companies listed on the stock market are representative of the economy as a whole, which might be proxied by high market capitalization and a large number of listed companies.

Table 2
Indicators of stock market development, size, liquidity, and legal origin

Countries	1988			1998							
	NGDP per cap (US\$)	Mkt cap (US\$ million)	Mkt cap percentage of GDP	NGDP per cap (US\$)	Mkt cap (US\$ million)	Mkt cap percentage of GDP	Turnover ratio	Listed cos	IPOs/ Pop	Legal origin	Anti-dir. rights
<i>Emerging markets</i>											
Argentina	4,047	2,025	2	9,518	45,332	13	28.8	130	0.20	FRA	4
Bangladesh	227	430	2	348	1,034	2	61.7	208	n.a.	n.a.	n.a.
Botswana	2,189	n.a.	n.a.	3,160	724	15	10.5	14	n.a.	n.a.	n.a.
Brazil	2,112	32,149	11	4,543	160,887	21	70.4	527	0.00	FRA	3
Bulgaria	5,113	n.a.	n.a.	1,356	992	8	2.4	998	n.a.	n.a.	n.a.
Chile	1,970	6,849	28	5,147	51,866	70	7.1	277	0.35	FRA	3
China, Mainland	278	n.a.	n.a.	772	231,322	24	130.1	853	n.a.	n.a.	n.a.
Colombia	1,297	1,145	3	2,404	13,357	15	9.4	163	0.05	FRA	n.a.
Cote d'Ivoire	948	437	4	699	1,818	16	2.5	35	n.a.	n.a.	n.a.
Czech Republic	5,246	n.a.	n.a.	5,169	12,045	23	38.2	261	n.a.	n.a.	n.a.
Ecuador	997	n.a.	n.a.	1,545	1,527	8	7.4	37	0.09	FRA	2
Egypt	1,698	1,760	2	1,297	24,381	30	22.2	861	n.a.	FRA	2
Ghana	368	n.a.	n.a.	374	1,384	19	4.8	21	n.a.	n.a.	n.a.
Greece	6,461	4,285	7	11,490	79,992	66	82.3	244	0.30	FRA	1
Hungary	2,995	n.a.	n.a.	4,734	14,028	29	111.3	55	n.a.	n.a.	n.a.
India	366	23,623	8	433	105,188	25	55.2	5,860	1.24	GBR	2
Indonesia	674	253	0	435	22,104	25	37.9	287	0.10	FRA	2
Israel	9,888	5,458	12	15,485	39,628	43	26.6	650	1.80	GBR	3
Jamaica	1,501	796	23	2,784	2,139	30	1.9	49	n.a.	n.a.	n.a.
Jordan	2,031	2,233	37	1,486	5,838	78	11.6	150	n.a.	FRA	1
Kenya	379	474	6	315	2,024	20	4.1	58	n.a.	GBR	3
Korea	4,336	94,238	52	6,694	114,593	37	176.2	748	0.02	DEU	2
Lithuania	9,074	n.a.	n.a.	2,843	1,074	10	16.0	60	n.a.	n.a.	n.a.
Malaysia	2,048	23,318	67	3,072	98,557	146	30.0	736	2.89	GBR	3
Mauritius	1,958	n.a.	n.a.	3,451	1,849	46	5.8	40	n.a.	n.a.	n.a.
Mexico	2,190	13,784	8	4,326	91,746	22	27.3	194	0.03	FRA	0
Morocco	948	446	2	1,259	1,849	5	9.9	53	n.a.	n.a.	n.a.

Nigeria	279	960	4	298	2,887	8	4.9	186	n.a.	GBR	3
Pakistan	378	2,460	6	458	5,418	8	111.1	773	n.a.	GBR	4
Peru	1,201	n.a.	n.a.	2,434	11,645	18	19.0	257	0.13	FRA	2
Philippines	645	4,280	11	875	35,314	55	30.0	221	0.27	FRA	4
Poland	1,724	n.a.	n.a.	3,840	20,461	14	54.7	198	n.a.	n.a.	n.a.
Portugal	5,033	7,172	15	11,618	62,954	58	93.4	135	0.50	FRA	2
Russia	5,990	n.a.	n.a.	1,921	20,598	7	9.1	237	n.a.	n.a.	n.a.
Saudi Arabia	5,076	n.a.	n.a.	6,164	42,563	34	26.9	74	n.a.	n.a.	n.a.
Slovenia	n.a.	n.a.	n.a.	10,044	2,450	12	34.4	28	n.a.	n.a.	n.a.
South Africa	2,488	126,094	143	2,553	170,252	146	29.1	668	0.05	GBR	4
Sri Lanka	421	471	7	825	1,705	11	14.8	233	0.11	GBR	2
Taiwan, Province of China	6,192	120,017	97	11,702	260,015	100	323.0	437	0.00	DEU	3
Thailand	1,131	8,811	14	1,906	34,903	30	71.0	418	0.56	GBR	3
Trinidad and Tobago	3,685	268	6	4,604	3,922	64	5.0	25	n.a.	n.a.	n.a.
Tunisia	1,299	612	6	2,172	2,268	11	8.2	38	n.a.	n.a.	n.a.
Turkey	1,654	1,135	1	2,995	33,646	17	144.9	277	0.05	FRA	2
Venezuela	3,291	1,816	3	4,055	7,587	8	13.6	94	0.00	FRA	1
Zimbabwe	851	774	10	446	1,310	23	10.1	67	n.a.	GBR	3
<i>Developed markets</i>											
Australia	15,666	138,283	53	19,249	874,283	240	51.9	1,162	n.a.	GBR	4
Austria	16,739	8,862	7	26,217	34,106	16	47.4	96	0.25	DEU	2
Belgium	15,630	58,920	38	25,050	245,657	98	28.9	146	0.30	FRA	0
Canada	18,510	241,880	49	19,725	543,394	91	67.0	1,384	4.93	GBR	4
Hong Kong (SAR)	10,358	74,377	128	26,510	343,394	206	54.4	658	5.16	GBR	0
Denmark	21,669	30,178	27	32,942	98,881	57	n.a.	242	1.80	SCA	3
Finland	20,995	30,179	29	24,524	154,518	123	53.0	129	0.60	SCA	2
France	17,262	244,833	25	24,685	991,484	69	68.7	711	0.17	FRA	2
Germany	15,479	251,777	21	26,148	1,093,962	51	144.9	741	0.08	DEU	1
Ireland	9,819	n.a.	n.a.	22,146	29,956	38	81.8	79	0.75	GBR	3
Italy	14,754	135,428	16	20,657	569,731	49	104.1	320	0.31	FRA	0
Japan	23,813	3,906,680	134	29,957	2,495,757	66	40.3	2,416	0.26	DEU	3

(continued on next page)

Table 2 (continued)

Countries	1988			1998					IPOs/ Pop	Legal origin	Anti-dir. rights
	NGDP per cap (US\$)	Mkt cap (US\$ million)	Mkt cap percentage of GDP	NGDP per cap (US\$)	Mkt cap (US\$ million)	Mkt cap percentage of GDP	Turnover ratio	Listed cos			
<i>Developed markets</i>											
Netherlands	15,688	113,565	49	24,099	603,182	159	70.7	212	0.66	FRA	2
New Zealand	13,150	13,163	30	13,974	89,373	170	56.2	135	0.66	GBR	4
Norway	23,245	14,332	15	32,917	56,285	39	69.4	236	4.50	SCA	3
Singapore	9,875	24,049	94	26,423	94,469	112	50.5	321	5.67	GBR	3
Spain	9,014	91,118	26	14,281	402,180	72	201.9	484	0.07	FRA	2
Sweden	21,503	100,083	55	25,479	278,707	123	73.9	258	1.66	SCA	2
Switzerland	28,290	140,527	75	36,669	689,199	262	100.8	232	n.a.	DEU	1
United Kingdom	14,563	771,206	93	23,497	2,374,273	171	53.4	2,399	2.01	GBR	4
United States	20,613	2,793,816	55	31,487	13,451,352	158	106.2	8,450	3.11	GBR	5

Sources: *Emerging Markets Factbook*, IFC; *International Financial Statistics*, IMF; and [LLSV \(1997\)](#).

Mkt cap is market capitalization. NGDP per cap is nominal GDP per capita. The turnover ratio is total value traded during the year divided by average market capitalization. Listed cos is the number of listed domestic companies. IPOs/Pop is the ratio of initial public offerings to population (in millions). GBR is English, FRA is French, DEU is German, and SCA is Scandinavian. Anti-dir. rights is the index of anti-director rights from [LLSV \(1997\)](#).

price changes provide information that managers will consider more relevant. Other country characteristics would not matter under these theories.

- Under the “financing” hypothesis, countries with well-developed financial markets as proxied by high market capitalization and a large number of listed domestic companies and initial public offerings can be expected to display a stronger link between stock returns and growth. In fact, the financing mechanism is more likely to operate in countries where it is easy to issue shares or take over other companies.
- Under the “stock market pressure on managers” hypothesis, countries in which managers are less protected from shareholders, that is — according to [La Porta et al. \(1997, henceforth LLSV\)](#) — countries whose stock market regulations are of English origin as opposed to Scandinavian, German, or French origin (in increasing order of managers’ protection), should display a stronger growth-returns association. In fact, if market sentiment brings about a decline in a firm’s stock price, its managers will be able to continue pursuing a strategy of high investment only where it would be difficult to fire them or to take over their firm.
- Under all of the theories above, stock market turnover, as a proxy for liquidity, would seem to be a potential determinant of the strength of the growth-returns association, whereas, controlling for the other financial development indicators, the degree of economic development or the distinction between “emerging” and “advanced” countries would not matter.

While the discussion above has focused on investment, which accounts for a very large portion of the variation in GDP growth, stock price developments can have a major impact on consumption as well, through their impact on wealth. One would expect this mechanism to be higher in countries where stocks constitute a large proportion of consumers’ portfolios — which may be proxied by market capitalization as a share of GDP.⁵

When turning to the data, this paper considers the growth of overall GDP rather than that of its components — such as investment or consumption — to analyze as large a sample of countries as possible. In fact, the number of countries with reliable data on the components of aggregate demand over a sufficiently long time span is limited, particularly in the case of emerging markets. As is well known, the comovement of investment and output (and of consumption and output) is extremely high in most countries (see, e.g., [Backus and Kehoe, 1992](#)). Moreover, existing studies have found that the link between investment and stock returns is similar to the link between overall output and stock returns ([Barro, 1990](#); [Fischer and Merton, 1984](#); [Mullins and Wadhvani, 1989](#)).

Testing of these theories has thus far relied almost exclusively on data from the United States. Using both aggregate and firm-level data from the United States, [Morck et al. \(1990\)](#) argue that the stock market is largely a “sideshow” which merely reflects changes in expected output growth. [Blanchard et al. \(1993\)](#) broadly support this view by showing that stock price movements independent of fundamentals have only a small impact on economic activity.

⁵ Data on the proportion of individuals’ portfolios that is accounted for by stocks are only available for a small number of advanced countries.

To my knowledge, the only existing study that exploits cross-country differences in the link between output growth and stock returns is by [Mullins and Wadhvani \(1989\)](#), who find that the link is stronger in the United States and the United Kingdom than in Japan or Germany. They attribute this difference to the greater possibility of takeovers, lower gearing ratios, more pervasive use of stock option schemes in managers' remuneration, and smaller role played by employees in decision making, that characterize the Anglo-Saxon countries.

A possible reason why previous research has not focused on the link between stock returns and output growth in emerging markets might be a presumption that stock markets are smaller and less liquid in emerging markets than in advanced countries. Although indicators such as the ratio of market capitalization to GDP are, on average, higher in advanced economies than emerging markets, several emerging markets score much higher on these indicators than most advanced economies ([Table 2](#)). Indeed, there is considerable variation in market capitalization and turnover ratios, the number of listed domestic companies and initial public offerings, and legal origin among both emerging and advanced countries. This valuable information is used in the next section.

3. Data and estimation

This section briefly describes the data and reports the estimation results.⁶ It analyzes the association between output growth and lagged stock returns in the countries in the sample, and considers whether its strength is the same for various groups of countries. The section begins by looking at whether there are differences between emerging markets and advanced countries (following the definitions of the Investment Finance Corporation, IFC): the objective is to find out whether stock returns could help predict output growth in emerging countries. The section then turns to possible differences among countries in terms of the variables identified by the theories discussed above: the focus here shifts toward exploring the mechanisms underlying the growth-returns association.

Data on real stock returns (obtained as the difference between nominal stock returns and consumer price inflation) and real GDP growth are available at an annual frequency for at least 22 years for 8 emerging countries and 17 advanced countries. At a quarterly frequency, data on real stock returns and real GDP growth are available for at least 10 years for 6 emerging countries and 18 advanced countries. To expand the number of emerging markets for which quarterly data are available, industrial production growth is also used, adding another 7 emerging countries to the sample.

This paper uses stock returns instead of a variable such as q , which might be considered to have more solid theoretical underpinnings as a determinant of output growth, owing to two practical considerations. First, data on the replacement cost of capital, which are needed to construct q , are not available for the sample of countries considered. Second, [Barro \(1990\)](#) and [Blanchard et al. \(1993\)](#) show that, in the United States, economic growth is more highly correlated with lagged stock returns than with a constructed q variable.

⁶ A more detailed description of the data, including variable definitions and sources, is provided in Appendix A. Additional robustness exercises are reported in [Mauro \(2000\)](#).

When testing whether the association between output growth and lagged stock returns is the same for different groups of countries, two estimation strategies are used: (i) individual-country regressions, followed by computing the average slope coefficient⁷ on lagged stock returns and, for univariate regressions, the average R^2 coefficients for each group; and (ii) panel regressions allowing for individual-country fixed effects but restricting the slope coefficients to be the same for all countries belonging to the same group.

3.1. *Emerging versus advanced countries*

Considering the 25 countries for which at least 20 annual observations are available, the univariate correlation between real economic growth and real stock returns (lagged by 1 year)⁸ is positive in all countries except India, and significantly positive in 5 out of 8 emerging markets and 10 out of 17 advanced countries (Table 3, “Univariate Regressions”).⁹ In the case of the advanced countries, the estimates are similar to those reported by previous studies. The slope coefficient typically ranges between 0.01 and 0.09, and amounts to 0.035 averaging over all countries in the sample. An increase in real stock returns by 10 percentage points is therefore typically associated with higher real economic growth by 0.35 percentage point. In 7 out of the 25 countries considered, the R^2 coefficient is 0.3 or higher. The average slope coefficient is slightly higher in emerging countries than advanced countries, though this result is reversed when the regression includes lagged growth as an additional independent variable (see below). Similarly, the R^2 coefficient is higher, on average, in emerging countries than in advanced countries (setting the R^2 coefficient to zero in the case of India).¹⁰

Following the existing literature, this paper focuses on the relationship between stock returns and economic growth by regressing real GDP growth in year t on GDP growth and stock returns in year $t - 1$. While this functional form is intuitively appealing, it might seem somewhat restrictive, raising a number of issues involved in the econometric estimation. In particular, both real GDP and an index of real stock prices have unit roots. In most countries considered, tests of whether there is a long-run relationship between real GDP and real stock prices (based on cointegration using the Johansen approach or on the estimation of parameters in an autoregressive distributive lag regression) fail to yield a clear-cut long-run relationship with sensible coefficients.

Controlling for lagged economic growth, real economic growth and real stock returns (lagged by 1 year) at the annual frequency are positively and significantly associated in 4

⁷ That is, the mean group estimator — an unbiased estimator of the coefficient for the group of countries even when the regressions include the lagged dependent variable (Pesaran and Smith, 1995).

⁸ As is well known from existing studies on advanced countries, output growth is most strongly correlated with stock returns lagged by 1 year (rather than contemporaneous stock returns or stock returns lagged by more than 1 year), and this specification is standard. If contemporaneous stock returns or stock returns lagged by more than 1 year are introduced as additional regressors, in virtually no country are they significantly different from zero, again confirming the results obtained by existing studies.

⁹ The results are similar when the years of the emerging market crisis, 1997–1998, are excluded.

¹⁰ Even though it is not always clear whether a given country ought to be classified as emerging or advanced, small changes in classification do not alter the group averages significantly. This result holds when the emerging market crisis years of 1997–1998 are excluded from the sample.

Table 3

Output growth and lagged stock returns, individual country regressions, 1971–1998

	Univariate regressions				Lagged dependent variable regressions				
	R^2	DW	Constant	Lagged stock returns	R^2	DW	Constant	Lagged growth	Lagged stock returns
<i>Emerging market countries</i>									
Argentina	0.03	1.72	1.81 (1.46)	0.011 (0.61)	0.03	1.84	1.68 (1.36)	0.077 (0.42)	0.012 (0.62)
Chile	0.39	1.67	3.67 (2.23)	0.085 (2.09)*	0.39	1.79	3.24 (2.66)	0.099 (0.66)	0.080 (1.74)
Greece	0.05	1.57	2.10 (4.91)	0.012 (1.15)	0.15	2.18	1.44 (3.64)	0.298 (1.76)	0.016 (1.24)
India	0.11	2.11	5.71 (13.58)	−0.041 (−2.12)	0.13	1.92	6.55 (10.46)	−0.161 (−1.35)	−0.043 (−2.22)
Korea	0.40	2.12	6.66 (10.19)	0.078 (3.03)**	0.40	2.09	7.12 (3.84)	−0.061 (−0.23)	0.081 (2.54)*
Mexico	0.13	1.40	2.79 (3.08)	0.035 (2.08)*	0.22	1.99	1.83 (1.65)	0.295 (1.16)	0.037 (2.41)*
Thailand	0.58	1.46	6.00 (8.05)	0.078 (3.15)**	0.77	1.45	0.27 (0.17)	0.817 (4.99)	0.045 (2.96)**
Zimbabwe	0.22	1.76	2.71 (2.79)	0.041 (3.53)**	0.22	1.90	2.55 (2.98)	0.063 (0.23)	0.040 (3.21)**
Average coefficient	0.22 ^a	...	3.93	0.037	0.29	...	3.08	0.178	0.033
<i>Advanced countries</i>									
Australia	0.32	2.79	2.95 (6.94)	0.092 (2.25)*	0.37	2.53	3.75 (4.85)	−0.230 (−1.64)	0.079 (2.29)*
Austria	0.05	1.89	2.64 (6.93)	0.020 (1.51)	0.06	2.13	2.35 (3.20)	0.105 (0.39)	0.019 (1.40)
Belgium	0.12	1.76	2.02 (4.83)	0.032 (1.75)	0.14	2.14	1.67 (2.55)	0.134 (0.58)	0.036 (1.77)
Canada	0.15	0.99	2.94 (4.34)	0.056 (1.97)*	0.26	1.65	1.84 (2.55)	0.342 (1.93)	0.056 (2.23)*
Denmark	0.14	1.56	2.06 (5.85)	0.026 (2.18)*	0.16	1.77	1.76 (3.41)	0.135 (0.82)	0.026 (2.41)*
France	0.02	1.20	2.36 (6.12)	0.008 (0.69)	0.22	2.10	1.26 (2.83)	0.416 (3.00)	0.016 (1.14)
Germany	0.08	1.42	2.07 (4.89)	0.022 (1.43)	0.18	1.86	1.27 (1.92)	0.341 (1.81)	0.030 (1.36)
Italy	0.02	1.71	2.37 (5.65)	0.008 (0.84)	0.02	1.87	2.14 (2.46)	0.091 (0.39)	0.009 (0.93)
Japan	0.33	1.27	3.16 (6.88)	0.053 (3.50)**	0.40	1.76	2.06 (2.21)	0.290 (1.42)	0.051 (3.25)**
Netherlands	0.13	0.88	2.26 (5.08)	0.026 (2.31)*	0.40	1.82	0.74 (1.28)	0.533 (3.08)	0.042 (4.29)**
Norway	0.06	0.90	3.46 (7.46)	0.012 (1.29)	0.25	1.56	1.92 (2.47)	0.436 (2.80)	0.012 (1.76)
Singapore	0.32	1.29	7.33 (12.53)	0.046 (3.31)**	0.47	1.76	4.12 (4.98)	0.405 (3.85)	0.039 (3.06)**
Spain	0.12	0.69	2.79 (4.95)	0.028 (3.43)**	0.59	2.04	0.82 (2.12)	0.664 (5.99)	0.032 (4.68)**

Table 3 (continued)

	Univariate regressions				Lagged dependent variable regressions				
	R^2	DW	Constant	Lagged stock returns	R^2	DW	Constant	Lagged growth	Lagged stock returns
<i>Advanced countries</i>									
Sweden	0.09	1.15	1.49 (3.28)	0.022 (2.01)*	0.26	1.52	0.67 (1.09)	0.395 (2.17)	0.032 (2.48)*
Switzerland	0.06	1.37	1.37 (2.14)	0.029 (1.21)	0.20	1.94	0.66 (0.97)	0.374 (2.72)	0.041 (1.42)
United Kingdom	0.20	1.51	1.97 (4.67)	0.036 (4.80)**	0.30	2.10	1.24 (2.11)	0.326 (1.80)	0.037 (3.99)**
United States	0.30	1.28	2.40 (5.68)	0.069 (3.79)**	0.34	1.52	1.84 (3.58)	0.200 (2.15)	0.071 (3.74)**
Average coefficient	0.15	...	2.68	0.034	0.27	...	1.77	0.292	0.037
<i>All countries</i>									
Average coefficient	0.17 ^a	...	3.08	0.035	0.28	...	2.19	0.255	0.036

Sources: MSCI web site; *Emerging Markets Factbook*, IFC; and *International Financial Statistics*, IMF.

Newey–West corrected t -statistics in parentheses. DW is the Durbin–Watson statistic. There are 22 observations for each emerging market and 28 observations for each advanced country.

^a The average R^2 is computed setting the R^2 in India to zero.

* Indicates that the coefficient is significant at the 5% level.

** Indicates that the coefficient is significant at the 1% level.

out of 8 emerging market countries and 10 out of 17 advanced countries (Table 3, “Lagged Dependent Variable Regressions”). Again, the only country displaying a negative association between growth and lagged stock returns is India. Using this specification, the coefficient on lagged stock returns is, on average, slightly higher in advanced countries than in emerging market countries.

The results obtained with individual-country regressions are confirmed by estimating panel regressions with individual-country intercepts but constraining the slope coefficients to be the same for all countries.¹¹ In a regression of real economic growth on lagged real stock returns, the estimated slope coefficient amounts to 0.034 and is highly significant (Table 4). When the slope coefficient is allowed to differ between emerging country and advanced country groups (though constrained to be the same for all countries within a group), the coefficient turns out to be higher in emerging countries (0.035) than in advanced countries (0.032), though not significantly so. The results are robust to changes in sample period, for example, excluding 1997–1998 (the years the Asian crisis) or the years up to 1977 for the advanced markets. The slope coefficient is somewhat higher in the first half of the sample period than in the second half, but not significantly so.

An advantage of panel estimation is that it makes it possible to use also the information available from those countries that have a smaller number of observations. When the estimation is conducted using all 41 countries for which at least 10 observations are

¹¹ This is a useful way of summarizing the information to be drawn from all countries, despite the variation seen in the individual country regressions and the significant slope differences across theoretically based groups of countries reported below.

Table 4

Panel estimation of relationship between output growth and lagged stock returns

Sample period	Estimation method	Countries	Number of observation	Lagged stock returns
1971–1998	Fixed effects	All 25 (20 + yr)	651	0.034 (10.1)
		Emerging 8	176	0.035 (5.37)
		Advanced 17	475	0.032 (8.34)
		All 41 (10 + yr)	846	0.030 (10.5)
	SURE	All 25 (20 + yr)	651	0.031 (20.53)
		Emerging 8	176	0.030 (5.83)
		Advanced 17	475	0.028 (12.89)
1977–1996	Fixed effects	All 25 (20 + yr)	499	0.027 (7.33)
		Emerging 8	160	0.029 (4.31)
		Advanced 17	339	0.024 (5.54)
		All 41 (10 + yr)	662	0.023 (7.32)
	SURE	Emerging 8	160	0.027 (5.34)
		Advanced 17	339	0.024 (13.05)

Sources: MSCI web site; *Emerging Markets Factbook*, IFC; *International Financial Statistics*, IMF.

All 25 (20 + yr) are the 25 countries (8 emerging and 17 advanced) with more than 20 years of data.

All 41 (10 + yr) are the 41 countries with more than 10 years of data. *t*-Statistics are reported in parentheses.

available, the coefficients are once again similar to those reported above. (The estimates are similar when an alternative minimum number of observations is used.) Estimation as a system of seemingly unrelated regressions (SURE) reduces the standard errors of the estimated coefficients considerably (especially for advanced countries).

Panel estimation shows that lagged stock returns remain significantly associated with output growth in both advanced and emerging countries when controlling for lagged values of other leading indicators, including real short-term interest rates, and the real growth rate of both narrow and broad money (Table 5). Narrow and broad money growth are used as controls because they have been found to be useful leading indicators for developing countries (Agenor et al., 2000) and there is a large literature on interest rates and the term structure as predictors of output (Estrella and Hardouvelis, 1991). The present paper confirms that low real interest rates and high real money growth tend to be followed by rapid output growth. These associations are significant, although narrow money has more predictive power in emerging countries, whereas broad money has more predictive

Table 5

Panel regressions of output growth on lagged stock returns and other leading indicators

Sample	Estimation technique	Number of observations	Stock returns	Stock returns times emerging market dummy	Short-term interest rate	Narrow money growth	Broad money growth
All 24 1977–1998	Least squares country dummies	503	0.0280 (7.26)		– 0.1263 (– 5.54)	0.0837 (5.97)	0.0339 (2.06)
7 Emerging 1977–1998	Least squares country dummies	130	0.0328 (3.89)		– 0.1216 (– 2.97)	0.1071 (3.81)	– 0.0002 (– 0.01)
17 Advanced 1977–1998	Least squares country dummies	373	0.0220 (5.54)		– 0.1387 (– 4.48)	0.0445 (2.78)	0.1147 (4.84)
All 24 1977–1998	Least squares country and year dummies	503	0.0303 (7.19)		– 0.1205 (– 5.13)	0.0766 (5.40)	0.0268 (1.61)
7 Emerging 1977–1998	Least squares country and year dummies	130	0.0306 (3.15)		– 0.1302 (– 3.13)	0.1245 (4.06)	– 0.0035 (– 0.11)
17 Advanced 1977–1998	Least squares country and year dummies	373	0.0271 (5.44)		– 0.1160 (– 2.75)	0.0180 (1.09)	0.0840 (3.51)
All 24 1977–1998	SURE country and year dummies	503	0.0285 (21.26)		– 0.1208 (– 15.11)	0.0735 (21.07)	0.0301 (5.95)
7 Emerging 1977–1998	SURE country and year dummies	130	0.0252 (3.68)		– 0.1277 (3.58)	0.1203 (4.93)	– 0.0008 (0.03)
17 Advanced 1977–1998	SURE country and year dummies	373	0.0269 (12.14)		– 0.1233 (7.47)	– 0.0008 (– 0.10)	0.0888 (9.59)
All 24 1977–1998	Least squares country dummies	503	0.0227 (4.36)	0.0109 (1.50)	– 0.1286 (– 5.64)	0.0814 (5.78)	0.0332 (2.02)
All 24 1977–1998	Least squares country and year dummies	503	0.0292 (4.76)	0.0021 (0.27)	– 0.1211 (– 5.13)	0.0762 (5.32)	0.0270 (1.61)
All 24 1977–1998	SURE country and year dummies	503	0.0269 (17.92)	0.0004 (1.58)	– 0.1217 (– 15.47)	0.0729 (21.16)	0.0301 (6.11)
All 24 1971–1998	SURE country and year dummies	589	0.0219 (9.98)	0.0058 (1.25)	– 0.1301 (– 8.78)	0.0617 (7.83)	0.0513 (5.17)
All 24 1977–1996	SURE country and year dummies	457	0.0258 (22.75)	0.0018 (0.75)	– 0.1192 (– 18.91)	0.0660 (21.33)	0.0315 (7.49)

Sources: MSCI web site; *Emerging Markets Factbook*, IFC; and *International Financial Statistics*, IMF.
All right-hand-side variables are in real terms and lagged by 1 year.

power in advanced countries.¹² The slope coefficient on lagged stock returns is, once again, similar in separate panels of advanced and emerging countries. In panels of all 24 countries¹³ for which data are available for at least 20 years, an interaction term of lagged stock returns times an emerging market dummy is not significant. The results are robust to changes in estimation technique (using individual-year dummies, and SURE) and sample period. On the whole, therefore, stock returns provide useful information to predict output growth, over and above that contained in other variables, not only in advanced countries but also in emerging countries, and the value of this information does not seem to be different in the two types of countries.

The evidence on the relationship between lagged stock returns and output growth at a quarterly frequency confirms the key findings obtained with annual data, though the proportion of countries in which the relationship is significant is lower among emerging market countries than among advanced countries. The correlation is analyzed between, on the right-hand side, real stock returns over the 12 months before time t and, on the left-hand side, the growth rate of real GDP or industrial production (depending on data availability) at different horizons, that is, in turn, between the quarter before t and the quarter following t ; the two quarters before t and the two quarters following t ; the three quarters before t and the three quarters following t ; and the four quarters before t and the four quarters following t .¹⁴ An appropriately lagged dependent variable and seasonal dummies are included in the estimates. In other words, the estimated equations are the following:

$$\begin{aligned} (400/j) \log \left[\left(\sum_{i=1}^j Y_{t-i} \right) / \sum_{i=1}^j (Y_{t-i}) \right] &= \alpha_0 + \alpha_1 \log[(S/P)_{t-j}/(S/P)_{t-j-4}] \\ &+ \beta (400/j) \log \left[\left(\sum_{i=1}^j Y_{t-j+i-1} \right) \right. \\ &\left. / \sum_{i=1}^j (Y_{t-j+i-1}) \right] + \varepsilon \end{aligned}$$

where Y is real GDP (or industrial production), S is the nominal stock price index, P is the consumer price index, and $j=1, \dots, 4$. Minus signs indicate lags. The correlation tends to be stronger, the longer the forecast horizon. Considering the case of $j=1$ (quarter-on-quarter growth), the correlation is positive and significant in 9 out of 18 advanced countries, and 2 out of 6 emerging markets, using real GDP; and 13 out of 18 advanced

¹² Controlling for narrow and broad money growth, private credit growth has a negative sign and is therefore omitted from the estimates. This does not affect any of the results related to stock returns.

¹³ Argentina is omitted from the sample in these estimates because its hyperinflation was mirrored in extremely negative measured real interest rates and money growth, which result in implausible coefficients on those variables in the panels. Nevertheless, none of the results related to the coefficient on lagged stock returns are affected by its inclusion in the sample.

¹⁴ This exercise is similar to that conducted by Estrella and Hardouvelis (1991).

countries, and 4 out of 13 emerging markets, using industrial production (Table 6).¹⁵ The results are similar controlling for other variables that other studies have found to be useful leading indicators of output, such as the real growth rate of credit to the private sector or broad money growth, and when the four quarterly real stock returns before time t are entered separately as right-hand-side variables.¹⁶

An alternative way of summarizing the association between output growth and stock returns, which is particularly useful in the case of quarterly data, is to estimate a vector autoregression system of output growth (quarter-on-quarter) and stock returns (over the previous quarter).¹⁷ The peak impact (usually reached two or three quarters after the initial shock) is positive in most countries in the sample; it is significant in 8 out of 15 advanced countries, but only in 3 out of 13 emerging markets.¹⁸

3.2. *Country characteristics and the strength of the growth-returns link*

This section generalizes the question of what type of country is more likely to display a strong association between stock returns and output growth. One may wonder whether the association might tend to be stronger, for example, in countries with higher market capitalization or turnover ratios, a greater number of listed domestic companies and initial public offerings, or a particular legal origin of the regulations governing the stock market. Given that these variables are all strongly correlated with each other, this question is addressed by estimating panel regressions including stock returns and stock returns interacted with each of market capitalization as a share of GDP, a dummy for English (or, alternatively, non-French) legal origin, the number of initial public offerings — following LLSV (1997) — as a share of population (in millions of inhabitants), and the logarithm of the number of listed domestic companies.

¹⁵ The fact that the proportion of countries where the relationship is significant is larger for advanced countries than for emerging countries is only partially due to the greater number of observations for advanced countries. When the advanced country regressions are estimated for the most recent 70 observations (a sample period roughly equivalent to that available for the emerging countries), the associations are significant in 8 out of 18 advanced countries, whether using real GDP or industrial production.

¹⁶ For example, controlling for private credit growth, the slope coefficient on lagged stock returns (over the previous year) is positive and significantly different from zero in 8 out of 18 advanced countries and 3 out of 6 emerging markets, using GDP; and in 13 out of 18 advanced countries and 3 out of 13 emerging markets, using industrial production (see Mauro, 2000).

¹⁷ This strategy is adopted by Hassapis and Kyprianos (2000), who find that the impulse responses of output growth to domestic stock return shocks (or the share of the variance of output growth that is attributable to stock return shocks) are not significant for Chile, Colombia, Mexico, and Peru, whereas they are significant for the United States. For these countries and methodology, the only difference worth mentioning between their results and mine is that I obtain a significant peak impact for Chile. As I have been able almost exactly to replicate these results of theirs using their same data sources, this difference may well derive from differences in the sample period, data frequency, and stock price indices used.

¹⁸ A similar pattern of results is obtained considering a variance decomposition 10 quarters after the shock, but the results tend to be less significant for both advanced and emerging countries. All these results are available from the author upon request.

Table 6
Output growth at different forecasting horizons, and lagged stock returns

Model	Industrial production									GDP								
	Regobs	<i>j</i> = 1		<i>j</i> = 2		<i>j</i> = 3		<i>j</i> = 4		Regobs	<i>j</i> = 1		<i>j</i> = 2		<i>j</i> = 3		<i>j</i> = 4	
		RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat		RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat	RSTK	<i>t</i> -Stat
		(-1)		(-2)		(-3)		(-4)			(-1)		(-2)		(-3)		(-4)	
<i>Emerging market countries</i>																		
Chile	89	0.124	2.27*	0.093	2.39*	0.077	2.02*	0.080	2.42*	74	0.108	2.57*	0.091	2.06*	0.078	1.83	0.061	1.61
Colombia	52	0.097	3.16**	0.090	3.36**	0.069	3.60**	0.045	3.42**									
Greece	89	0.002	0.06	0.007	0.30	0.013	0.72	0.008	0.52									
India	89	− 0.055	− 1.48	− 0.032	− 1.36	− 0.028	− 2.08	− 0.021	− 1.81									
Jordan	75	0.058	0.41	0.056	0.64	0.025	0.41	0.003	0.07									
Korea	75	0.060	0.98	0.074	1.55	0.087	2.36*	0.094	2.81**	89	0.141	1.90	0.143	4.29**	0.088	4.23**	0.064	3.90**
Malaysia	52	0.188	3.05**	0.161	4.02**	0.137	2.83**	0.094	2.61**									
Mexico	89	0.036	1.54	0.037	1.91	0.039	2.16*	0.039	2.16*	75	0.026	1.37	0.022	1.71	0.023	2.06*	0.022	1.92
Pakistan	52	0.035	0.46	0.030	0.76	0.023	0.82	0.016	0.67									
Philippines	53	0.058	0.84	0.063	1.27	0.064	1.42	0.059	1.57	52	0.031	1.44	0.028	2.05*	0.019	1.99*	0.019	2.27*
Portugal	43	0.015	0.33	0.035	1.36	0.056	2.67**	0.069	4.09**	43	0.060	2.26*	0.062	3.71**	0.069	6.18**	0.059	6.70**
Turkey	45	0.008	0.23	− 0.013	− 0.51	− 0.006	− 0.30	− 0.006	− 0.42	44	0.000	0.00	0.008	0.25	0.005	0.27	− 0.002	− 0.13
Zimbabwe	71	0.106	2.21*	0.086	1.91	0.086	2.13*	0.067	2.48*									
<i>Advanced countries</i>																		
Australia	113	0.097	2.71**	0.091	3.21**	0.096	3.90**	0.086	3.58**	113	0.232	1.57	0.140	2.18*	0.096	2.50*	0.078	2.63**
Austria	112	0.055	1.89	0.040	1.60	0.044	1.76	0.048	2.00*	90	0.014	0.84	0.028	2.14*	0.019	2.16*	0.018	2.25*

Canada	113	0.164	3.48**	0.153	3.49**	0.161	4.51**	0.146	4.81**	113	0.049	2.63**	0.054	2.81**	0.058	3.23**	0.056	3.30**
Denmark	113	0.162	3.14**	0.119	3.09**	0.109	3.14**	0.098	3.37**	112	0.021	0.74	0.018	0.86	0.024	1.19	0.026	1.29
Finland	41	0.089	3.07**	0.097	4.54**	0.117	5.76**	0.128	6.14**	39	0.099	3.61**	0.064	3.82**	0.074	4.17**	0.080	4.16**
France	112	0.087	2.34*	0.071	2.15*	0.068	2.36*	0.059	2.39*	113	0.018	1.62	0.015	1.50	0.016	1.76	0.015	1.73
Germany	112	0.089	3.93**	0.071	2.96**	0.075	3.30**	0.071	3.49**	46	0.010	0.24	0.019	0.76	0.008	0.49	0.011	0.92
Italy	112	0.052	1.31	0.042	1.37	0.041	1.43	0.037	1.47	110	0.014	1.57	0.018	1.57	0.020	1.64	0.016	1.39
Japan	113	0.165	3.15**	0.132	3.19**	0.146	3.49**	0.144	3.68**	112	0.052	2.07*	0.036	2.23*	0.033	2.34*	0.035	2.52*
Netherlands	112	0.126	3.13**	0.101	3.01**	0.101	3.33**	0.088	3.45**	86	0.054	2.41*	0.051	2.75**	0.044	2.79**	0.044	2.75**
New Zealand	41	0.080	1.40	0.120	2.14*	0.110	3.45**	0.084	3.03**	41	0.118	2.24*	0.100	3.04**	0.087	3.75**	0.077	4.23**
Norway	112	0.075	3.15**	0.046	2.96**	0.037	2.64**	0.036	2.72**	112	0.042	1.49	0.030	1.63	0.017	1.37	0.013	1.41
Singapore	113	0.124	3.11**	0.133	4.15**	0.131	5.36**	0.120	6.00**	53	0.102	3.55**	0.037	2.81**	0.031	2.80**	0.032	2.99**
Spain	112	0.011	1.63	0.024	2.41*	0.034	2.98**	0.042	3.34**	112	0.012	3.33**	0.019	4.04**	0.025	4.59**	0.028	4.72**
Sweden	112	0.105	3.00**	0.064	3.66**	0.068	4.07**	0.067	4.00**	112	0.041	1.86	0.038	2.68**	0.031	3.32**	0.030	3.74**
Switzerland	113	0.152	1.55	0.114	1.66	0.097	1.77	0.089	1.96*	113	0.048	1.32	0.049	1.64	0.052	1.77	0.051	1.78
United Kingdom	113	0.083	3.09**	0.083	3.22**	0.070	3.91**	0.060	4.63**	113	0.061	3.66**	0.048	4.37**	0.042	3.96**	0.041	4.18**
United States	113	0.174	3.15**	0.163	3.46**	0.160	4.38**	0.146	4.39**	113	0.071	3.35**	0.062	3.57**	0.065	4.01**	0.061	3.90**

Sources: MSCI web site; *Emerging Markets Factbook*, IFC; and *International Financial Statistics*, IMF.

The *t*-statistics are Newey–West corrected. Seasonal dummies and one lag of the dependent variable are included in the set of independent variables. Regobs is the number of observations in the regression. RSTK is the real stock return, and the number in brackets indicates the log at the quarterly frequency.

* Indicates that the coefficient is significant at the 5% level.

** Indicates that the coefficient is significant at the 1% level.

Table 7
Panel regression results with interaction terms, 1971–1998

Estimation technique	Fixed effects, no year dummies				Fixed effects with year dummies				SURE with year dummies						
Stock returns	0.0337 (10.1)	0.0208 (4.3)	0.0285 (1.2)	0.0206 (3.9)	0.0125 (1.7)	0.0318 (8.7)	0.0226 (4.7)	0.0177 (0.8)	0.0223 (4.1)	0.0171 (2.4)	0.0284 (17.7)	0.0181 (8.2)	0.0071 (0.7)	0.0165 (6.2)	0.0112 (3.6)
Stock returns times market capitalization/GDP		0.0232 (2.2)	0.0523 (3.4)	0.0213 (2.2)	0.0129 (1.3)		0.0197 (1.9)	0.0471 (3.2)	0.0195 (2.0)	0.0119 (1.2)		0.0239 (4.3)	0.0438 (5.9)	0.0273 (4.5)	0.0193 (3.4)
Stock returns times English origin dummy		0.0155 (2.0)	0.0135 (1.1)	−0.0040 (−0.5)			0.0102 (1.4)	0.0076 (0.7)	−0.0087 (−1.2)			0.0127 (2.9)	0.0044 (0.8)	−0.0065 (−1.3)	
Stock returns times number of IPOs/Population			−0.0079 (−2.8)					−0.0078 (−2.8)					−0.0055 (−5.3)		
Stock returns times log of listed domestic companies			−0.0016 (−0.4)					0.0008 (0.2)					0.0017 (1.0)		
Stock returns times index of anti-director rights					0.0045 (1.6)					0.0023 (0.8)					0.0029 (2.1)
Real interest rates			−0.1374 (−6.7)	−0.1359 (−6.6)					−0.1194 (−5.2)	−0.1193 (−5.2)				−0.1239 (−8.4)	−0.1269 (−8.6)
Real narrow money growth			0.0872 (6.3)	0.0870 (6.3)					0.0773 (5.5)	0.0769 (5.5)				0.0634 (8.1)	0.0625 (8.0)
Real broad money growth			0.0525 (3.3)	0.0499 (3.1)					0.0380 (2.3)	0.0359 (2.2)				0.0501 (5.2)	0.0492 (5.1)
Number of observations	651	651	573	589	589	651	651	573	589	589	651	651	573	589	589

Sources: *Emerging Markets Factbook*, IFC; *International Financial Statistics*, IMF; and *LLSV* (1997).

Left-hand-side variable: output growth. All right-hand-side variables are lagged by 1 year.

All panel regressions include individual country dummies. *t*-Statistics are reported in brackets.

The results suggest that the interaction terms with market capitalization and — in some specifications — English legal origin are positive and significant (Table 7).¹⁹ (The interaction term with the number of IPOs takes a negative and significant value, but this result is reversed if this is the only interaction term in the regression). Taken literally, for exposition purposes, the panel results (SURE with year dummies) that use market capitalization and English origin as interaction terms suggest that the magnitude of a country's slope coefficient in the returns-growth regressions would approximately double if a country were to double its market capitalization to GDP ratio, and it would increase by almost three quarters if a country of non-English origin were to move to English-style stock market regulations.

The interaction term on market capitalization is fairly robust to specification changes, such as introducing short-term interest rates and broad and narrow money growth as additional controls, whereas the individual significance of the interaction term on English origin is not robust to this specification change. In fact, when these controls are introduced in the regression, the coefficient on the interaction term on English origin changes sign.²⁰ An alternative to the English origin dummy is the anti-director rights' index of LLSV (1997), which has the advantage of being more precisely measured (it takes a value between 0 and 5), but also the disadvantage that it is not clear whether an increase from — say — 0 to 1 has the same meaning as an increase from 4 to 5. When the interaction term with the index of anti-director rights is used as an alternative, the estimated coefficient is positive, but its significance is not robust to specification changes. At the same time, when controlling for the index of anti-director rights, even market capitalization loses its significance in some specifications. On the whole, there is tentative evidence that legal origin matters, but the estimated coefficient on the interaction term with market capitalization is more robust to changes in specification.

3.3. Interpretation and possible extensions

Relating these results to the theories presented above, a possible interpretation might include the following.

- The finding that several countries with very low market capitalization (such as Mexico and Zimbabwe) display an economically and statistically significant growth-returns association suggests that the association may reflect in large part the mechanism emphasized by the “passive predictor” hypothesis: news about next year's growth is

¹⁹ Preliminary tests of whether countries have a different slope coefficient depending on whether they have — say — a high or a low turnover ratio are presented in Mauro (2000). As the turnover ratio is the only variable that turns out insignificant in such tests, it is omitted from the final specification in Table 7. Owing to data limitations, the 1998 observations of these variables are used for the turnover ratio and the number of listed domestic companies. The number of IPOs is from LLSV (1997) and refers to June 1995–July 1996. All of these variables seem to be strongly autocorrelated, although there are exceptions in which countries' relative positions changed over time.

²⁰ Using the market capitalization to GDP ratio in 1998 instead of the 1980–1998 average leads the estimated coefficients on the interaction terms with both market capitalization and English origin to be always positive, and more significant, in all specifications.

revealed during the course of this year, and is reflected in this year's stock price changes. In fact, on a priori grounds, economic causality would seem unlikely to run from stock returns to output growth in countries with such low market capitalization.

- The result that market capitalization is a useful predictor of whether a country will tend to have a strong growth-returns association is consistent with all theories outlined above. Those who believe in the “passive predictor” or the “active informant” hypotheses might argue that as market capitalization increases, companies listed on the stock market become more and more closely representative of the economy as a whole. By contrast, proponents of the “financing” or “stock market pressure on managers” mechanisms might point out that market capitalization is a proxy for a well-developed stock market where takeovers and share issues can easily be undertaken.
- The fact that, controlling for market capitalization, neither the number of IPOs nor the number of listed domestic companies seem to predict the strength of the growth-returns association implies that there is no evidence in favor of the “financing” hypothesis. At the same time, it might be argued that the number of IPOs and domestic companies are not good enough proxies for the variable that one would ultimately want to capture, namely the ease of takeovers and new share issues.
- The result that, controlling for other country characteristics, legal origin seems to matter for the growth-returns association would identify a role for the “stock market pressure on managers” hypothesis. When market sentiment turns against a firm, managers operating in countries where protection from stockholders is low will cut their investment plans to protect themselves against the possibility of a takeover or of being fired. However, as this result is not robust to specification changes, the evidence is tentative.

To shed further light on these issues, one question that seems worth exploring in future research is whether worldwide stock returns (or stock returns in key partner countries) have better predictive power than domestic stock returns do for domestic output growth, and the theoretical implications of any such empirical link that might be identified. In an initial effort to address this question, [Hassapis and Kyprianos \(2000\)](#) report results consistent with the view that US stock returns may have better predictive power for industrial production growth in Colombia and Mexico (but not in Chile and Peru) than domestic stock returns do. However, in regressions of output growth on lags of both domestic and worldwide (or US) stock returns using the data analyzed in this paper, I find that domestic stock returns turn out to be significant with the expected sign, unlike worldwide (or US) stock returns, in the vast majority of countries where stock returns have significant predictive power.²¹ Further research will have to explore the robustness of these findings in detail.

4. Concluding remarks

This paper has shown that there is a positive and significant correlation between output growth and lagged stock returns in several (though by no means all) countries, including both advanced countries with highly developed stock markets and developing countries

²¹ The results are available upon request from the author.

with emerging but still relatively undeveloped stock markets. The presence of this association in a variety of countries at different stages of economic and financial development suggests that the relationship is fairly robust, and that developments in stock prices should be taken into account in forecasting output, in both advanced and emerging countries. The result that the correlation is significant even in some countries with relatively small market capitalization also seems to lend support to the view that the correlation between output growth and stock returns is, to a considerable extent, due to a simple causal link from news about output growth to stock returns. Nevertheless, it is important to bear in mind that any empirically observed relationship whose underlying causal basis is not entirely understood may easily cease to exist in the future.

The paper finds that a number of characteristics of countries' stock markets are helpful, though far from perfect, predictors of the strength of the correlation between output growth and stock returns. Countries with a high market capitalization to GDP ratio, a large number of listed domestic companies and initial public offerings, and English origin of the regulations governing the stock market tend to display a significantly stronger correlation. In particular, both market capitalization and — less robustly — English origin are found to play an individually significant role. The significant and relatively robust role of stock market capitalization is not surprising, as it is to be expected on the basis of several theories. The role of English legal origin is more intriguing, as it would tend to support the view that “stock market pressure on managers” mechanisms help explain the link between output growth and lagged stock returns, but the evidence is tentative.

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Appendix A. Data Description

Data on real stock returns (obtained as the difference between nominal stock returns and consumer price inflation) and real GDP growth are available at an annual frequency over 1970–1998 for Australia, Austria, Belgium, Canada, Denmark, France, Germany (omitting 1991 — the observation affected by unification), Italy, Japan, the Netherlands, Norway, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States; 1976–1998 for Argentina, Chile, Greece, India, Korea, Mexico, Thailand, and Zimbabwe; 1979–1998 for Jordan; 1980–1998 for Brazil; 1985–1998 for Colombia, Malaysia, Nigeria, Pakistan, the Philippines, Taiwan Province of China, and Venezuela; 1987–1998 for Portugal; 1988–1998 for Finland, Indonesia, Ireland, New Zealand, and Turkey.

At a quarterly frequency, data on real stock returns and real GDP growth are available since 1970 for Australia, Austria, Canada, France, Germany (omitting the first quarter of 1991 — the observation affected by unification), Italy, Japan, Norway, Spain, Sweden, Switzerland, the United Kingdom, and the United States; since 1976 for Korea; since 1977 for the Netherlands; since 1980 for Chile and Mexico; since 1984 for Singapore; since 1985 for the Philippines; since 1986 for Portugal; since 1987 for Denmark and Turkey; and since 1988 for Finland, Malaysia, and New Zealand. To expand the number of emerging markets for which quarterly data are available, industrial production growth is also used. This is available for at least 40 observations for Colombia, Greece, India, Jordan, Pakistan, and Zimbabwe in addition to the countries listed above.

All stock price indices are in local currency. The stock price indices for the advanced countries are drawn from the website of Morgan Stanley Capital International (MSCI) (www.ms_cidata.com). They include dividend payments. For the emerging markets, the stock price indices are the IFC Global Indices, which cover 70–75% of total market capitalization and are drawn from the data bank of the International Finance Corporation (IFC). For a number of countries, both MSCI and IFC indices are available; in those cases, the correlation between the price indices is very high (always above 0.95).

The data on stock market capitalization, turnover ratio, and number of listed domestic companies are drawn from the *International Finance Emerging Markets Factbook* and the International Federation of Stock Exchanges, FIBV (www.fibv.com). The data on the number of initial public offerings of equity (in July 1995–June 1996) as a ratio of a country's population (in millions of inhabitants), and on the stock market regulations' legal origin are from LLSV (1997).

The data on real GDP, industrial production, consumer prices, narrow money, broad money, and private credit, are drawn from the International Monetary Fund's *International Financial Statistics*. The data set on industrial production is complemented by data from the OECD Analytical Database. The inflation rate, the growth rate of real GDP, and rates of return on stocks are obtained as log-differences.

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